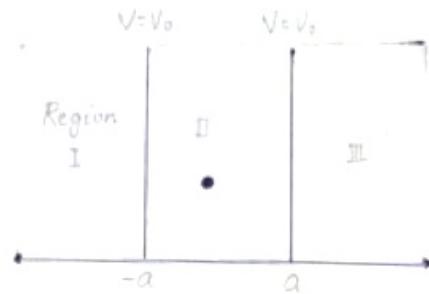


Finite Potential well

- Two boundaries where a finite force prevents a particle, moving freely from escaping is called the finite square well.



- The Hamiltonian is given by

$$\boxed{H = T + V}$$

where, $T \rightarrow$ K.E & $V \rightarrow$ P.E

- Since for particle in the finite s. well

$$V(x) = \begin{cases} 0, & -a \leq x \leq a \\ V_0, & a < x < -a \end{cases}$$

K.E $\rightarrow P^2/2m$ in the form of momentum

K.E $\rightarrow \frac{1}{2}mv^2$ " " " general term

P.E $\rightarrow 0$ in the form of free particle

P.E $\rightarrow V_0$ in the form of finite term

For Region I and III

- Region-I ($x < -a$) and Region-III ($x > a$): In this region potential energy is non-zero ($V = V_0$). So the schrodinger wave eqⁿ becomes.

$$\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + (E - V_0) \psi = 0$$

$$\text{or } -\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} = (E - V_0) \psi$$

$$\text{where, } k^2 = -\frac{2m}{\hbar^2} (E - V_0)$$

$$\boxed{\frac{\partial^2 \psi}{\partial x^2} = k^2 \psi}$$

To get simple differential eqⁿ

The general solutions to this differential equation in region I and III are

$$\psi_1(x) = Ae^{-kx} + Be^{kx}$$

$$\psi_1(x) = Be^{kx} \quad \text{--- (1)}$$

The first term in ψ_1 becomes non-physical for $x \rightarrow -\infty$, so the solⁿ in this region is -

$$\psi_3(x) = Ce^{-kx} + De^{kx}$$

$$\psi_3(x) = Ce^{-kx} \quad \text{--- (2)}$$

Similarly, the second term in ψ_3 becomes non-physical for $x \rightarrow +\infty$. So the solⁿ in this region is -

For Region - II

→ Region II ($-a < x < a$): In this region potential energy is zero.

$$\text{i.e. } V(x) = 0$$

So, the schrodinger wave eqⁿ becomes.

$$-\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} = E\psi$$

$$\text{or } \frac{\partial^2 \psi}{\partial x^2} = -\frac{2mE}{\hbar^2} \psi \quad \text{where, } L^2 = \frac{2mE}{\hbar^2}$$

$$\boxed{\frac{\partial^2 \psi}{\partial x^2} = -L^2 \psi}$$

To get simple differential eqⁿ

The general solⁿ to this differential equation is

$$\psi_2(x) = F \sin Lx + G \cos Lx \quad \text{--- (3)}$$

Boundary conditions

- At $x = -a$ the boundary conditions are:

$$\Psi_1(-a) = \Psi_2(-a) ; \quad \frac{\partial \Psi_1(-a)}{\partial x} = \frac{\partial \Psi_2(-a)}{\partial x}$$

- Similarly at $x = +a$ the boundary conditions are:

$$\Psi_2(a) = \Psi_3(a) ; \quad \frac{\partial \Psi_2(a)}{\partial x} = \frac{\partial \Psi_3(a)}{\partial x}$$

- These equations have two types of solutions; symmetric and anti-symmetric.
- The anti-symmetric solⁿ for boundary $x = -a$ using eqⁿ ① and eqⁿ ⑤ we get.

$$\underline{B e^{-ka} = -F \sin la} ; \quad \underline{B k e^{-ka} = -F l \cos la}$$

Dividing the both eqⁿ.

$$\underline{k = -l \cot la}$$

- The symmetric solⁿ for boundary $x = a$ using eqⁿ ② and eqⁿ ③ we get

$$\underline{C e^{-ka} = G \cos la} ; \quad \underline{-C k e^{-ka} = -G l \sin la}$$

Dividing the both eqⁿ.

$$\underline{k = l \tan la}$$

In the form of Energy

The constant k is given by $k^2 = -\frac{2m}{\hbar^2} (E - V_0)$

On Rearranging the eqⁿ we get energy

$$E = V_0 - \frac{\hbar^2 k^2}{2m}$$

Since $k = l \tan l a$ for symmetric and $k = -l \cot l a$ for antisymmetric, So we get two energy values.

$$E = V_0 - \frac{\hbar^2 l^2 \tan^2 l a}{2m}$$

$$E = V_0 - \frac{\hbar^2 l^2 \cot^2 l a}{2m}$$

Note:- If we put, $l^2 = \frac{2mE}{\hbar^2}$ we get another two values of energy